Utilizing Solar Thermal Energy for Hostel Cooking

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Abstract

The course project aims to find out the feasibility of implementing a solar thermal energy cooking system in Hostel-17 of IIT Bombay. Our aim in this report is to study the techno-economic feasibility of the system using the calculated values of DNI and the number of Scheffler Dishes required which was explored in earlier parts of the report. Finally, we compared the energy and costs saved using the proposed system compared to the conventional method of LPG Cooking.

Keywords: Scheffler Dish, Direct Normal Irradiance

1. Introduction

Our Hostel-17 mess currently uses LPG as a fuel source for cooking purposes, It caters to around 1300 students daily and has a current energy requirement of 5900 MJ. We have chosen a Scheffler Dish for our system which will collect the solar energy required for the cooking system. This solar energy will turn liquid water into steam which will be further used for cooking and act as an alternative to LPG Fuel.

By using we have done a cost analysis of the system to find out the total cost of the system and calculate the total energy savings

The system below is a visual depiction of the system with the streams

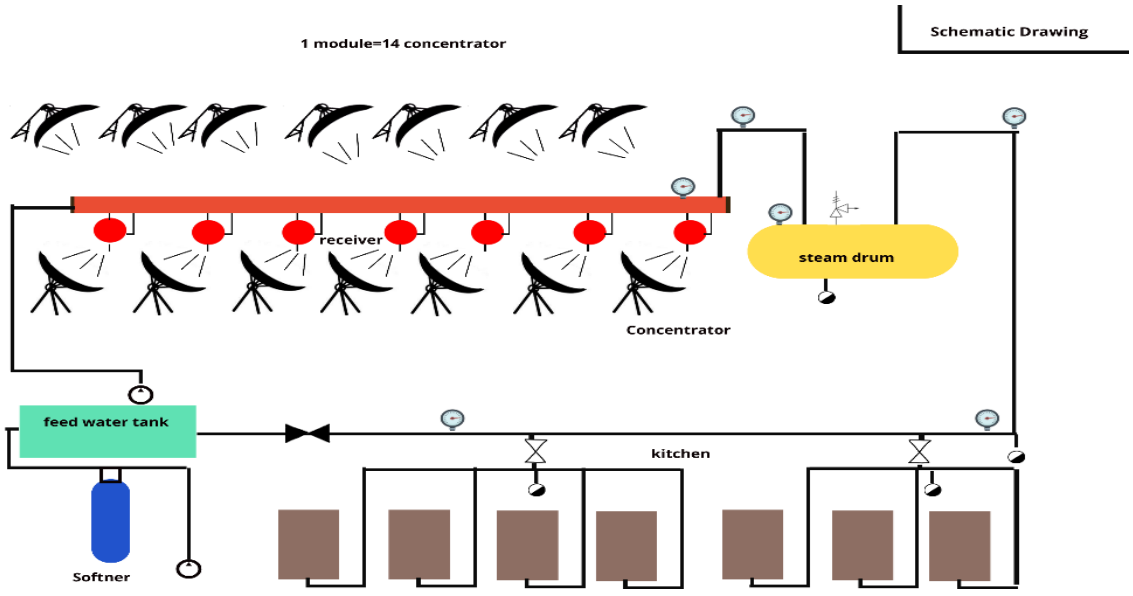


Figure1: Schematic of the System

2. Plant configuration

2.1 Scheffler Dish

Scheffler dish is a small lateral section of a paraboloid, which concentrates the sun's radiation over a fixed focus, with automatic single-axis tracking.

Working Principle

The Scheffler dish is made of several flat-shaped mirrors or reflective films that are mounted on a structural frame. The dish rotates about a north-south axis parallel to the earth's axis to track the sun's movement. The axis of rotation passes through the center of gravity of the reflector and that's how the reflector always maintains its gravitational equilibrium. The Scheffler reflector also performs a change in inclination angle while staying directed to the sun, in order to obtain a sharp focal point. Focus is fixed at the distance of the focal length of the paraboloid along the axis of the paraboloid. The receiver at a fixed location captures the concentrated heat and transfers it to water/thermic fluid to generate hot water/hot thermic fluid or highpressure steam.

Subsystem

The receiving subsystem collects solar radiation in the solar field from parabola-shaped reflecting collectors and transfers it to working fluids such as water or oil. The tracking subsystem acts as a support to the receiving subsystem. It maximizes the solar energy captured by moving solar collectors towards the sun for a complete day. The circulation subsystem carries fluid and transfers the heat received by it to the end-use application.

Components

- **Collector -** The collector of Scheffler Dish is an assembly of flat-shaped solar-grade glass mirrors or Aluminium mirror reflectors arranged on a structural steel framework. The mirrors are arranged to give a paraboloidal shape and reflect the incident solar radiation to a common point, i.e. the receiver.

- **Receiver -** The receiver of the Scheffler dish is placed at the focus of the dish so as to capture the incident solar radiation and transfer it to the thermal medium used in the system. The critical component is generally a dome-type absorber fixed onto the structural steel frame. The receiver works on the thermo siphon principle in which cold water sinks down from the header and hot water/steam moves up to the header.

- **Header Pipe –**
  - Storage tank contains water in the lower half and steam in the upper half
  - Works as a buffer system, i.e. it collects and stores steam generated by the Scheffler system, which can be utilized as per the requirement

- **Dish and Receiver Support**
- **Tracking System**
- **Piping and Insulation**
- **Pumps**
- **Instrumentation and Safety mechanisms**
- **Heat Storage System-** We will be utilising large boiler type steel tank for storing steam since htf will be high pressured steam/water.

2.2 Applications

Application Type - Residential.

Food consumption pattern over a day and its breakdown:

Energy Calculations per day:

Gas cylinders (14.2kg) - 9 Nos.
LPG has a calorific value of 46.1 MJ/kg.

14.2 cylinder has an energy content of 654.6 MJ of heat.
We have a total of 9 Cylinders which corresponds to total energy used per day = 5900MJ

System Characteristics	
Heat Transfer Medium	Water
Temperature	150 – 200 ˚C
Pressure	3 – 7 bar
LPG Energy usage per Day	5900 MJ
Energy required by steam for Cooking Rice, curry and Dal, Steaming milk and Water	1436 MJ

Table 1: System Characteristics

2.3 DNI Value

Determining Solar Radiation value for Sizing the system

· Hourly averages are calculated at the mid of every hour given in the weather file
· The monthly average of every hour is calculated for every month of the year
· The total daily average energy for a month is obtained by summing the average energy values calculated for each hour throughout the month
· The daily solar radiation value in kWh/m2 is determined by averaging the monthly average energy day over one year

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Time of day	Hourly DNI(W/m^2) Jan	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov	Dec	Annual Averaged	
2	0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
3	1.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
4	2.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
5	3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
6	4.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
7	5.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
8	6.5	0	0	0	0	0	0.166667	0	0	0	0	0	0	0.0138889	
9	7.5	0.645161	49.9643	112	153.933	162.387	74.4333	36.2581	69.2258	144.7	154.484	143.867	20.8065	93.5587	
10	8.5	422.161	440.357	448.129	417.8	338.613	201.933	97.4516	162.323	294.367	374.548	438.9	348.355	332.078	
11	9.5	618.645	630.714	610.161	562.1	451.806	262.5	138.935	241.968	333.167	496	587.233	527.226	455.038	
12	10.5	703.161	736.357	702.323	651.2	531.452	339.133	200.677	258.806	300.533	553.129	647.6	626.71	520.923	
13	11.5	729.194	784.357	756.484	680.7	570	379.433	157.677	219.581	319.333	569.161	679.233	682.032	543.932	
14	12.5	739.258	808.357	780.29	681.133	616.581	308.8	168.419	257.258	296.467	568.581	701.467	694.581	551.766	
15	13.5	749.194	804.25	766.903	687.2	623.645	298.2	163.677	167.548	353.167	520.935	651.333	686.194	539.354	
16	14.5	729.387	785.143	722.387	665.8	599.935	263.067	146.258	182.484	396.133	520.419	617	641.774	522.482	
17	15.5	647.355	723.429	685.516	615.033	534.129	246.2	144.194	146.194	355.7	472.097	497.867	553.71	468.452	
18	16.5	521.548	583.857	563.161	481.8	427.935	218.9	98.3871	136.581	260.8	327.387	342.3	367.903	360.88	
19	17.5	170.097	342.964	324.613	262	205	88.9333	60.7419	51.8065	116.233	60.8387	11.8333	10.4839	142.129	
20	18.5	0	0	0	0	3.35484	12.1	1.58065	0.419355	0	0	0	0	1.45457	
21	19.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
22	20.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
23	21.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
24	22.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
25	23.5	0	0	0	0	0	0	0	0	0	0	0	0	0	
26															
27	Total Energy per day	6030.645161	6689.7493	6471.967	5858.699	5064.83784	2693.799267	1414.25635	1894.194655	3170.6	4617.5797	5318.6333	5159.7754	4532.061414	kWh/m2/Day
28	No. of Days	31	28	31	30	31	30	31	31	30	31	30	31		
29		186950	187312.9804	200630.977	175760.97	157009.973	80813.97801	43841.94685	58720.03431	95118	143144.9707	159558.999	159953.0374		
30															
31											Annual Energy intercepted		1648815.867	Wh/m^2	
32													1.648815867	MWh/m^2	

Table x: Average DNI Calculation for design considerations

3. Performance analysis model

3.1 Efficiency of Dish

Optical Efficiency

$\eta_{opt,rec}$= ρ × η_{cos}. × Absorptivity of receiver

ρ = Reflectivity of the mirror
η_{cos} = Cosine efficiency

Concentration Ratio - $\frac{11.65}{0.1294}$ = 90.3091
$\eta_{optical}$ = $cos\ cos\ \theta$ × reflectivity of the collector x absorptivity of the receiver.
$cos\ cos\ \theta$ is calculated for May 17 at 12.30 p.m.
N = 137 | β = ϕ = 19.07600˚ | δ = 19.2636250 ˚ | ω = -1.1445660 ˚ | $\gamma = 0$ ˚

On substitution,
$cos\ cos\ \theta$ = 0.943822
Reflectivity of collector = 0.9 | Absorptivity of receiver = 0.7

$\eta_{optical}$ = 0.943822 × 0.92 ×0.7 = 0.607821

Reciever End Efficiency

Heat loss at the receiver end = Loss due to convection + Loss due to radiation
The aperture area of receiver(A_r) = 0.1294 m^2
Convective Heat Transfer Coefficient(*h*) =25 W/(m^2 ·K)
T_r=473 K : T_a=301 K

$T_{sky} = 0.00553 \times (T_a^{1.5}) + 10 = 298K$ **(Swinbank's equation)**
$q_{loss} = \{h \times A_r \times (T_r - T_a)\} + \{\sigma \times \varepsilon \times A_r \times (T_r^4 - T_{sky}^4)\}$
$q_{loss} = \{25 \times A_r \times (473 - 301)\} + \{5.67 \times 10^{-8} \times 0.2 \times A_r \times (473^4 - 298^4)\}$
$q_{loss} = 4778.1 \times 0.1294$
$q_{loss} = 618.29$ W
Aperture area of reflector(A_{ref}) = 11.65 m^2

Energy absorbed by a receiver: Average DNI $\times A_{ref}$ × optical efficiency
= 665 × 11.65 × 0.6078 = 4708.9 W

Heat loss due to convection and radiation = 618.29 W.
$\eta_{the\text{-}rcv}$ = (Energy absorbed-Heat loss)/Energy absorbed
=(4708.9-618.29)/4708.9
$\eta_{the\text{-}rcv}$ = 0.8687

Overall thermal Efficiency = $\eta_{optical} \times \eta_{rcv}$
=0.607 × 0.868
$\boldsymbol{\eta_{overall}}$ = 0.528.

3.2 Energy Savings

We are only considering hours of radiation with value above 200 W/m^2
According to our DNI calculations, the net energy assimilated after making the mentioned consideration of 200 W/m^2 is 1.6488 MWh/m^2.
The optical efficiency of the reflector and the efficiency of the receiver came out to be 0.607 and 0.868 respectively.
Hence, the net energy that is made useful to us is given by;
Useful Energy = Net energy Intercepted in a year x Optical Efficiency x Efficiency of the receiver.
1.6488 MWh/m^2 x 0.607 x 0.868 x 16 x 14 = 194.591 MWh per yr

4. Results and discussion

This section can be divided into sustainability analysis and techno-economic analysis. Sustainability analysis deals with the amouth of LPG fuel saved and tonnes of CO2 emission saved.In economic Analysis, we deal about Simple payback period, ROI, Benefit to cost ratio and comparison with a research paper to get a overview of other solar thermal systems.

4.1 Sustainability Analysis

- LPG cylinder - 14.2 kg propane (C_3H_8).
- $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ + Heat
- LPG equivalent of steam produced to achieve cooking = 1436 MJ / 655 MJ -> 2.2 Cylinders.
- Cylinders Saved by incorporating Scheffler based Cooking = 2.2/0.6 =3.67 cylinders per day i.e 52.06 Kg of LPG saved per day.

CO2 calculations

C3H8 + 5O2 ----------> 3CO2 + 4H2O + Heat.
44 gm of propane has 36 gm of C.

In a cylinder of 14.2 kg propane we have 11.61 Kg of C.
C + O2 ——————> CO2
1 mole of C produce one mole of CO2 i.e 12 gm of C gives 44gm of CO2
Thus 11.61 kg of C gives 42.6kg of CO2. CO2 saved per cylinder=42.6kg.
42.6 x 3.67 = 156.34 Kg of CO2 emission is saved.
Hence per year = (156.34x 365)/1000 =57.06 Tonnes

4.2 Economical Analysis of the Scheffler cooking system

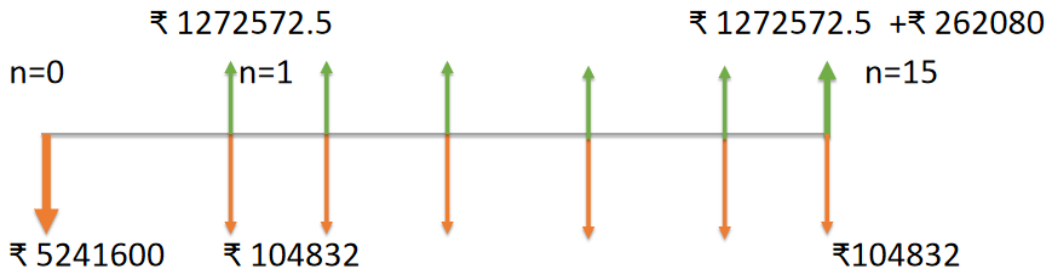
As per govt of India single axis tracking based Scheffler system costs 18,000 per m2 .
Cost of one Scheffler dish of 16m2 = 16 x 18,000= 288,000 Rs.

Cost of retrofitting existing kitchen and adding suitable accessories = 30% of capital cost = 374400 per dish
Total cost of system of 14 dishes = 5241600 Rs
Fuel saving cost per yr
Cost of LPG cylinder= 950 Rs
Fuel saving cost per yr = 950 x 365 x 3.67 = 1272572.5 Rs

SPP =Capital Investment/ Annual Savings = 4.11 yrs

ROI = (1/SPP)x100 =24.33%

Expected life of Scheffler dish = 15 yrs
Annual cost of O&M = 2% of capital cost =104832 Rs
Annual escalation in cost of LPG = 1.31 times
Discount rate = 10%
Salvage value = 5% of capital cost =262080 Rs



Net Present Value of the project
$=\sum NPV\ of\ fuel\ saving + NPV\ Salvage - NPV\ O\&M - Capital\ Cost$
NPV of fuel saving= 9679287.61 Rs
NPV of O&M = 797360.5 Rs
NPV of Salvage=62739.86 Rs
NPV of Project= 9679287.61+ 62739.86 -797360.5 -5241600
=3703066.09 Rs

Benefit to Cost Ratio =
(NPV fuel saving + NPV salvage value)/(Capital cost + NPV O&M)
=1.613

IRR = Discount rate at which NPV is zero(taking n=15), on solving
IRR=21.07%

Table 5
Estimated numbers of Scheffler dish, capital investment required and breakeven price of diesel of Scheffler dish based SSC system for institutional kitchens in India.

City	Number of Scheffler dish			Capital investment ($C_{o,sc}$) ('000 INR)			Breakeven unit price of diesel (*BEP*) (INR/litre)		
	Small (200 persons)	*Medium (500 persons)*	*Large (1000 persons)*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*
Jaipur (hot and dry)	2	5	10	861	1822	3213	59	50	44
Delhi (composite)	4	9	17	1518	2948	4960	86	74	66
Shimla (cold and cloudy)	3	6	12	1200	2116	3730	62	55	48

Table 6
Measures of financial performance of Scheffler dish based SSC system for small, medium and large sized institutional kitchens in India.

City	Discounted payback period (years)			Net Present Value (NPV_e) (INR '000)			Benefit to Cost ratio $(B/C)_e$			Internal rate of Return (IRR_e) (%)		
	Small	*Medium*	*Large*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*
Jaipur (hot and dry)	9	7	6	347	1247	2991	1.35	1.59	1.80	15.62	19.19	22.11
Delhi (composite)	21	14	11	−138	228	1132	0.92	1.07	1.20	8.59	11.15	13.29
Shimla (cold and cloudy)	15	9	8	375	1077	2730	1.27	1.44	1.64	14.42	17.00	19.75

5. Conclusion

Our study shed light on the benefits of utilising a Scheffler based solar cooking system which will fulfill partial energy requirement of Hostel-17 mess. In our Sustainability analysis it was measured that, we save around 57.06 Tonnes of Carbon dioxide a year.

Performing similar analysis on the economics of the system, it was observed that the ROI came out to be 24.33%.

Nomenclature

Abbreviations

LAT	Local Apparent Time
IST	Indian Standard Time
DNI	Direct Normal Irradiance

Symbols

I_g	Global Radiation
I_b	Beam Radiation
I_d	Diffused Radiation
I_T	Radiation on tilt surface
I_{bn}	Beam Normal Radiation
θ	Angle of Incidence
θ_z	Zenith Angle
δ	Declination Angle
$\emptyset$	Latitude
γ	Surface Azimuthal Angle
ρ	Albedo constant
A_i	Anisotropic Index

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